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Palaeomagnetic investigations of the Palaeozoic circum-Sowie Góry Mountains ophiolitic belt in the Sudetes, Poland

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SUMMARY

The investigated material comprises rocks of the Palaeozoic ophiolitic assemblage exposed around the Sowie Góry Mountains Massif in the Sudetes in the form of the three units: Ślęża, Ząbkowice and Nowa Ruda. The radiometric age (U–Pb) of the rocks, stated as 420 ∓ 20 Ma, places them in the Silurian. A palaeomagnetic study reveals the presence of four groups of palaeomagnetic poles. One of these is derived from an isolated component representing Permian overprinting. The second remanence component fits with the Silurian results from Baltica and may therefore be regarded as being of Silurian age. According to this interpretation, the ophiolitic complex would not have been involved in tectonic action after Silurian times. On the other hand, this component does not agree with the results of the neighbouring Bohemian Massif but is consistent with Late Carboniferous data of the European Variscans. In this case the other two components should be interpreted as Early- to Mid-Carboniferous overprints. This interpretation implies important strike-slip movements. In order to exclude one of these hypotheses, more data are still needed.

Key words: ophiolite, palaeomagnetism, remagnetization, Silurian, Sudetes, Variscides

INTRODUCTION

The nature of the geotectonic evolution of the Variscan orogenic belt of Europe remains an unsolved problem. Although modern ideas are based on plate tectonics, the proposed models diverge strongly due to the lack of convincing evidence for any oceans within the Variscan Belt. There are several models for the geotectonic evolution of the Variscides. Some authors (Matte 1991; Ziegler 1986) favour the existence of ocean basins inside the Variscan Belt. Matte (1991) proposes the model of convergence and collision of Gondwana and Laurentia–Baltica after closure of the Iapetus and Rheic basins to the north and the Galicia–Massif Central basin to the south.

The model of Franke (1989) for Central Europe is based on the concept of 'exotic' or 'suspect terranes' with narrow ocean basins in between. Neugebauer (1989) presents a different hypothesis. According to him, there were no oceans inside the Variscan area and the only oceans involved in the Caledonian, Variscan and Alleghenian orogenies were the Iapetus and its branch, the Tornquist Sea. The Variscan Belt was part of the continental margin of Gondwana which became compressed and consolidated after the subduction of the adjacent ocean. Badham (1982) suggests that the Variscan orogen was formed due to oblique collision of the European and African continental plates. These oblique interactions resulted in strike-slip faulting, bimodal volcanism and ophiolitic assemblages. The idea of allochthonous terranes derived from the continental margin colliding obliquely with the margin of the Baltica plate and accompanying strike-slip faults was also expressed by Martinez-Catalan (1990). This short review of geological hypotheses concerning the European Variscides shows the importance of two problems: the verification of oceanic basins and the possibility of strike-slip movements before and during the orogeny.

According to palaeomagnetic data, Variscan Europe resulted from the collision of the Laurentia, Gondwana, Baltica, Avalonia and Armorica plates. 'Armorica', introduced by Van der Voo on the basis of palaeomagnetic

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evidence, extends from the Iberian Meseta, through the Armorican Massif, the Central Massif and the Vosges, to the Bohemian Massif (Van der Voo 1988). Perroud, Calza & Khattach (1991) and Ruffet, Perroud & Feroud (1992) suggested that Armorica was part of the northern margin of Gondwana during the whole Ordovician–Silurian time interval and became separated from it sometime during the Devonian or Lower–Middle Carboniferous.

Tait, Bachtadse & Soffel (1994) showed that the Bohemian Massif and other parts of Armorica were positioned at high southern latitudes during Early–Middle Ordovician times. On the other hand, Torsvik *et al.* (1993) proposed pre-Wenlock times for the collision of Baltica and Eastern Avalonia (southern Britain) to form Balonia, and latitudinal closure of the Tornquist Sea and the British segment of the Iapetus Ocean.

Among the palaeomagnetic data for Variscan Europe, the results representing the Silurian–Lower Carboniferous time-span are the most scarce and uncertain. For this reason, the authors decided to undertake a palaeomagnetic study of rocks of the Silurian ophiolitic assemblages from the Sudetes.

GEOLOGIAL SETTING

The Sudetes form the north-eastern extremity of the Bohemian Massif, which lies in the eastern part of the European Variscan Belt (Fig. 1). They consist of a variety of small domains exhibiting different lithostratigraphic, tectonic and metamorphic characteristics.

Low-grade metamorphosed or even unmetamorphosed mafic to ultramafic complexes, which can be considered as remnants of oceanic crust, surround the Precambrian allochthonous gneissic massif of the Sowie Góry Mountains (Fig. 1). The massif itself is cut by the Marginal Sudetic Fault (MSF) striking NW–SE. The largest unit studied, Ślęża (SL), lies to the north of the Sowie Góry Mountains, and the smallest unit, Ząbkowice (ZA), is to the south-east.

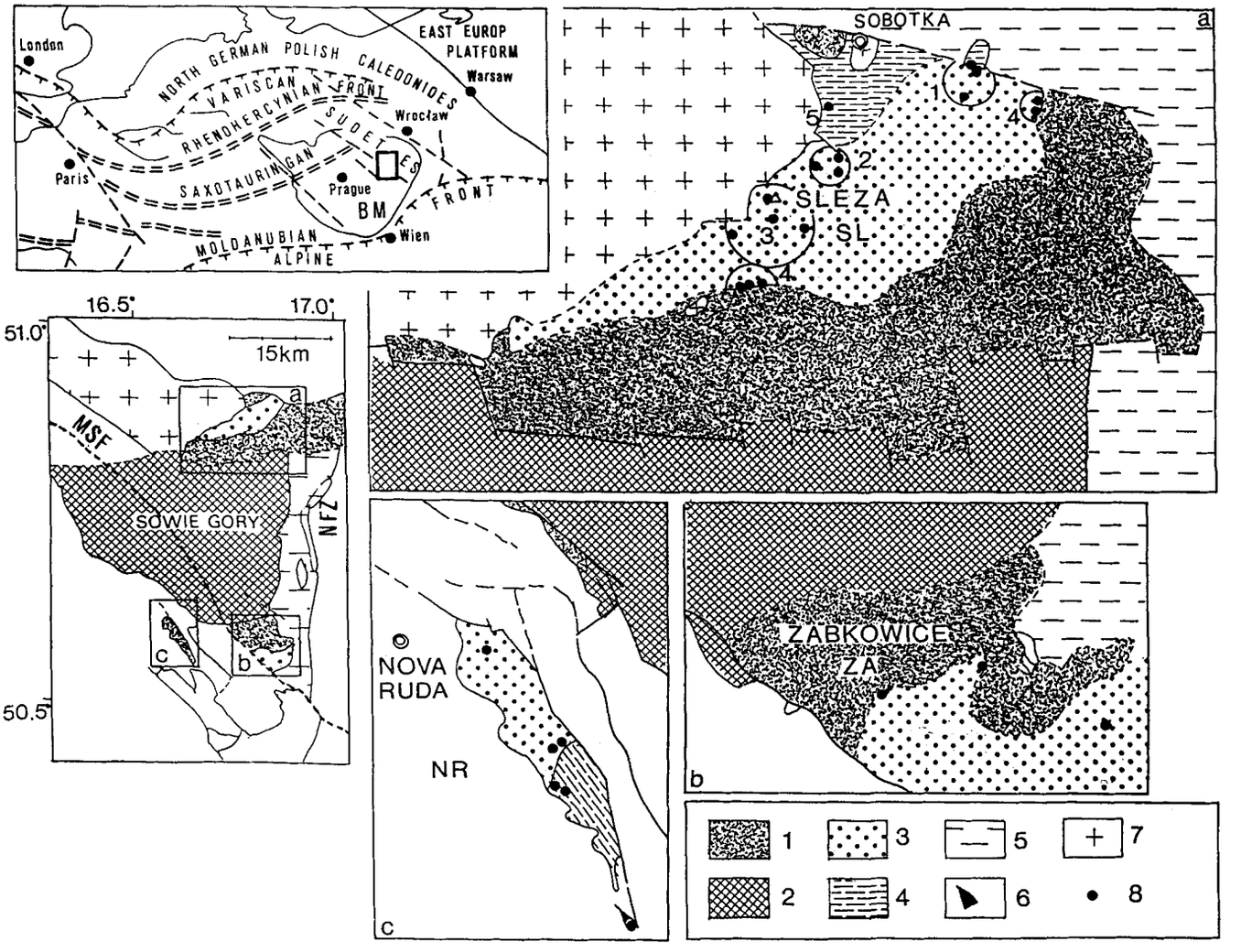


Figure 1. Geological sketch maps of the region investigated: situation of the Sudetes in the Variscan Belt, and position of the studied ophiolitic units around the Sowie Góry Mountains. BM—Bohemian massif; MSF—Marginal Sudetic Fault; NFZ—Niemcza Fault Zone. a Ślęża unit (SL), b Ząbkowice unit (ZA), c Nowa Ruda unit (NR). Full circles denote sample locations, and the numbers in a indicate zones of the Ślęża unit as listed in Table 1. 1—serpentinite; 2—Sowie Góry gneisses; 3—gabbro; 4—diabase; 5—Niemcza Fault Zone; 6—pillow lavas; 7—granitoids; 8—sampling sites.

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Both are in contact with the Sowie Mountain gneisses and are north of the MSF. ZA is close to the Niemcza Fault Zone (NFZ), which determines the eastern edge of the Sowie Góry. The third unit, Nowa Ruda (NR) lies on the southern side of the Sowie Góry and to the south of the MSF (Fig. 1). According to Żelaźniewicz & Cwojdziński (1994), the faults probably became active in the Carboniferous and have been rejuvenated several times since that time. The Alpine orogeny did not affect the Sudetes.

There is no consistent radiometric age for the rocks in question. The age obtained by Pin, Majerowicz & Wojciehowska (1988) with the Nd–Sm method is 353 ∓ 21 Ma for SL and 351 ∓ 16 Ma for NR, placing them in the latest Devonian–earliest Carboniferous. More recently, Oliver, Corfu & Krogh (1993) reported that the U–Pb ages for zircons and titanites from the same regions were 420 ∓ 20 Ma. This result places the investigated rocks into the Silurian. These results agree with previous estimates of the age of the Sudetic ophiolites expressed by geologists on the basis of stratigraphic studies. It is possible that the Nd–Sm age determines the time of the main metamorphic event.

Geochemical investigations of the rocks from the three regions being studied suggest that they belong to the normal MORB affinity (Pin *et al.* 1988).

Figures 1(a), (b) and (c) present the sampling sites within the three complexes under study. The SL complex (Fig. 1a) is composed of gabbroic massif and some amphibolites, both overlain by older, highly deformed and serpentinized peridotites. In some places, gabbro is in contact with diabase dykes. Palaeomagnetic studies were performed only on gabbros and diabases. The gabbroic massif is regarded by Jamrozik (1989a,b) as a stratified intrusion rather than the fragment of the ophiolitic complex *sensu stricto* suggested by Majerowicz & Pin (1989). To the north-east, the SL gabbros are in contact with the Upper Carboniferous granitoid massif.

According to Jamrozik (1989a), palaeohorizons of the crystallization zones of SL gabbros retained their primary horizontal (or subhorizontal) position. This conclusion is supported by the horizontal position of the exposed zone enriched in Fe–Ti minerals. Gabbros are younger than the serpentinites and cut them with multiple undisturbed apophyses.

According to Żelaźniewicz & Cwojdziński (1994), the Ślęża ophiolite consists almost exclusively of unfoliated and statically metamorphosed basic and ultrabasic rocks. The ophiolite is in direct contact with the Sowie Góry Mountain gneisses but the contact itself is poorly exposed. the cited authors write: '... it is evident ... [that the ophiolite–Sowie Góry contacts] ... are not ductile but dominantly cold and brittle, of normal fault type ... and obviously related to the late uplift of both domains'.

There is no evidence concerning tectonic tilting of the two other ophiolitic units ZA and NR. One of the authors (LJ) assumes that they retained their primary horizontal position as in the case of the SL unit.

Aleksandrowski (1994), in his analysis of Sudetian tectonics, suggests the possibility of strike-slip dextral displacements of about 300 km along the Main Sudetic Fault in Late Devonian (?)–Early Carboniferous times in the Western Sudetes. In the Eastern Sudetes the displacements did not exceed several tens of kilometres. Unfortunately, such small displacements cannot be identified by palaeomagnetic methods.

The 16 sampling sites in the SL unit are distributed over five zones (Fig. 1a). Two gabbroic zones, numbered 1 and 2 are situated in the Fe–Ti mineralized part of the massif. The mineralization is of magmatic origin and comprises ilmeno–magnetite minerals with a predominance of ilmenite. Sulphides are present as accessory minerals in the gabbros and as the main magnetic carriers in diabases (zone 5). From the SL massif, 68 gabbroic and five diabase hand samples were collected. The lithology and magnetic mineralogy is presented in Table 1.

The ZA complex (Fig. 1b) comprises mostly serpentinized ultrabasic rocks and gabbroids. According to Dziedzicowa

Table 1. Characteristics of rocks from Ślęża, Ząbkowice and Nowa Ruda

Unit/zone	Lithology	Magnetic mineralogy		Susceptibility $k \times 10^{-4}$ SI	NRM mA m^{-1}
		Optical	Thermomagn.		
Ślęża SL					
SL1	gabbro, diabase	altered ilm–mgt	magnetite	7–20	10^3–10^6
Fe–Ti enriched		pyrrhotite			
SL2	gabbro	altered ilm–mgt	magnetite	5–1200	10^3–10^7
Fe–Ti enriched		pyrrhotite			
SL3	gabbro	pyrite, pyrrh.	pyrrh., mgt	2–5	10^3–10^6
top zone					
SL4	gabbro	magnetite, pyrrh.	mgt, pyrrh.	2–6	10^2–10^4
edge zone	amp–pyr gabbro			250–1000	10^6–10^7
SL5	diabase	altered ilm–mgt	pyrrhotite	8–40	10^3–10^8
dyke					
Ząbkowice ZA	gabbro	very small amount of ore minerals	magnetite	4	10^2–10^4
Nowa Ruda NR					
	pillow lava	Fe–ox, sulphides	pyrrh., mgt	5–7	10^2–10^3
	diabase	Fe–ox, sulphides	mgt, pyrrh.	5–200	10^2–10^4
	gabbros	altered ilm–mgt sulphides	mgt, pyrrh.	2–10	10^3–10^5

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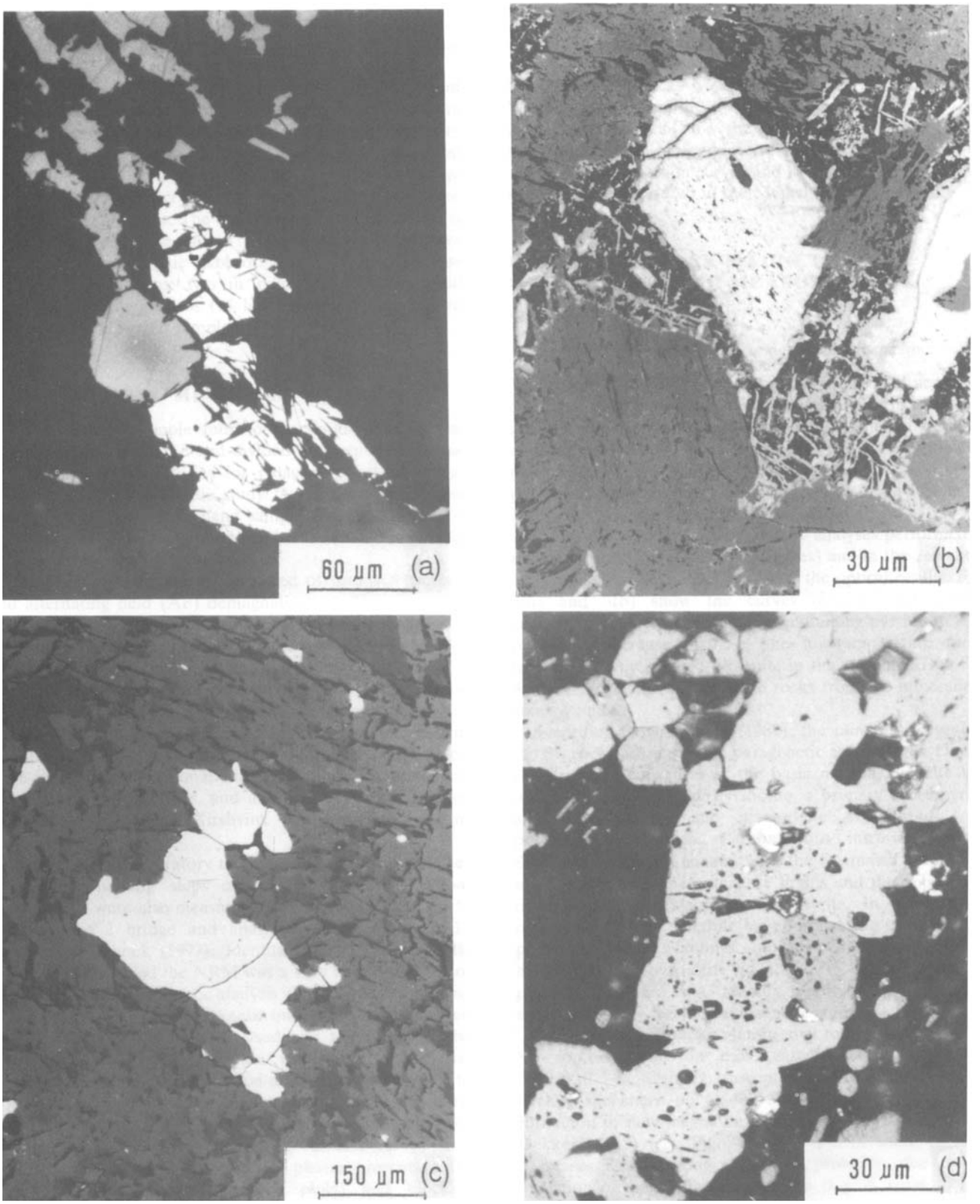


Figure 2. Examples of polished sections. (a) SL amphibole-pyroxene gabbro with a magnetite grain; (b) SL gabbro with a highly altered titanomagnetite grain; (c) SL gabbro with an unchanged ilmenite grain; (d) diabase with pyrrhotite grains.

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(1989), some of the gabbros were subjected to dynamic recrystallization either during the final stages of magmatic crystallization or after magma emplacement, when igneous and metamorphic events overlapped. 13 hand samples of gabbroids were collected from three sites. At one site recrystallized rocks outcropped. The lithology and magnetic mineralogy are summarized in Table 1.

The NR unit (Fig. 1c) comprises gabbros, diabases and pillow lavas. The latter rock type outcrops uniquely within this unit. According to Borkowska *et al.* (1989), the majority of the gabbroic rocks show post-magmatic alteration processes. They crystallized in the temperature range 1260–900°C, and their crystal fractionation took place under low-pressure conditions. The diabases are probably products of solidification of the parent basic magma at shallower depths than comagmatic gabbros. The NR complex was sampled at nine sites: two in pillow lavas (nine hand samples), three in gabbros (15 hand samples) and four in diabase dykes (18 hand samples)—see Table 1.

EXPERIMENTAL METHODS

From each hand sample four to six standard cylindrical specimens were prepared and divided between the Warsaw and Strasbourg laboratories for palaeomagnetic investigation. For the rock-magnetic study two to three specimens were prepared from each hand sample. The rock-magnetic investigations were performed in the Warsaw palaeomagnetic laboratory and in Munich.

The palaeomagnetic study consisted of standard thermal and alternating field (AF) demagnetization. The maximum demagnetization temperatures were 700°C, whilst the maximum alternating fields were in the range 0.1–0.14 T. All prepared specimens were demagnetized.

The NRM measurements were carried out on the IR-4 magnetometer and the 2G cryogenic magnetometer in Warsaw and on a modified Digico spinner magnetometer in Strasbourg. Demagnetization results from each specimen were plotted on orthogonal Zijderveld diagrams as well as on equal-area projections, and analysed with least-squares methods of line fitting (Kirshvink 1980) independently in both laboratories.

In the Warsaw laboratory the magnetic susceptibility after consecutive heating steps and its anisotropy at room temperature were also measured. Measurements were made with a KLY-2 bridge and analysed with the Anizo 11 program of Jelinek (1977). Identification of the magnetic minerals that carried the NRM was a very important part of this study. Thermomagnetic analysis with respect to blocking temperatures (T_b) of the magnetic minerals was also carried out in Warsaw. The method used was to determine the decay curves of the saturation remanence, I_{rs}, during continuous heating in a non-magnetic space up to 650–700°C. The same procedure was repeated twice—first for the fresh sample and then for the same sample after the first heating and cooling. The two I_{rs} versus T curves show blocking temperatures of magnetic phases present in the fresh rock (first curve in I_{rs}–T plots) and blocking temperatures of phases present in the rock after heating (second curve in I_{rs}–T plots).

In the Munich laboratory, thermomagnetic analysis of the same material was performed in air with a Curie balance in a field of about 0.5 T, to obtain the Curie temperatures (T_c) of the magnetic minerals. The saturation magnetization I_s versus T curves were registered both during heating in air to 700°C and during cooling. For some samples the I_s–T curves were measured for temperatures ranging from −196 to 700°C. In addition, an optical study of polished sections of the investigated rocks was performed in both Munich and Warsaw. The purpose of this was to identify the ore minerals responsible for the magnetic properties, their transformation phases and interrelations.

The IRM acquisition curves and hysteresis loops for some samples were measured in the Munich laboratory.

RESULTS OF ROCK-MAGNETIC AND PALAEOMAGNETIC MEASUREMENTS

Sleża complex

Magnetic mineralogy and rock-magnetic characteristics

The optical study of polished and thin sections of nearly all collected samples revealed abundances of ilmenite, altered ilmenite and magnetite (or Ti–magnetite), small amounts of unaltered magnetite (or Ti–magnetite) and some sulphides, partly altered. The diabase dykes were mainly made up of pyrrhotite. In some cases primary pyrrhotite inclusions in ilmenite grains were visible (Figs 2a–d).

The results of the thermomagnetic analyses performed in the saturation field (I_s versus T curves) and in the zero field (I_{rs} versus T curves) agree well with the optical results. Figs 3(a) and 3(b) show the curves obtained for gabbro containing magnetite and diabase containing pyrrhotite. The important decrease of I_s in the low-temperature range indicates the presence of ilmenite in the specimen (see Fig. 3). It is especially visible in the rocks from the mineralized zones 1 and 2.

According to Niśkiewicz (1989), the minerals present in the SL rocks belong to five paragenetic assemblages. During the early crystallization of the basic magma, ilmenite and magnetite were formed. Pyrrhotite, a primary chalcopyrite, and also valleriit and arsenpyrite crystallized from post-magmatic liquids. Carboniferous intrusion of the Strzegom granite in contact with the SL massif resulted in martitization of some magnetite grains and the appearance of a secondary chalcopyrite and pyrite. In recent times covelline and some goethite have formed due to weathering processes. These findings indicate that some of the magnetite and pyrrhotite identified in the SL rocks is primary. We did not identify goethite in either the mineralogical or the thermomagnetic studies.

Values of susceptibility measured at room temperature for SL specimens are in most cases in the range 100–2000×10^{-6} SI. Usually, susceptibility begins to rise after heating specimens up to 400–500°C, probably due to the formation of new magnetite from sulphides (Kruczyk 1983, Dekkers 1988) (see Fig. 4, curve 'a'). In some cases, it decreases after heating to 600°C, probably, due to the oxidation of magnetite to haematite (Fig. 4, curve 'b'). In the samples with pyrrhotite as the main magnetic carrier, the susceptibility decreases above 300–400°C and then begins to rise again (Fig. 4, curve 'c'), probably due to the transforma-

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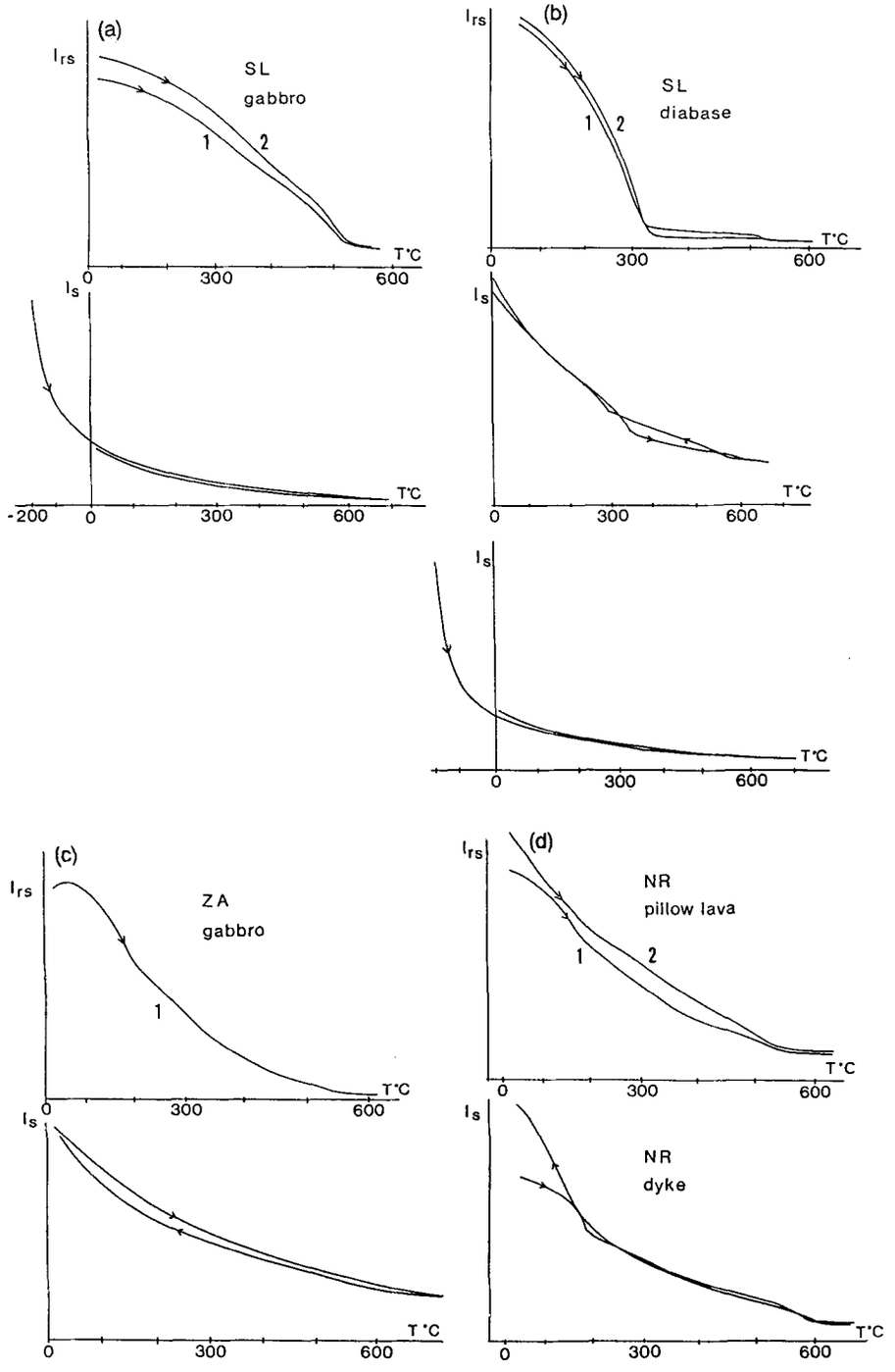


Figure 3. Thermomagnetic curves. (a) I_{rs}–T curves for SL gabbro from zone 1 (upper figure) and I_s–T curves in the range −196–700°C (lower figure) for the same sample. (b) I_{rs}–T curves for SL diabase from zone 5 (upper figure), I_s–T curves in the range of 20–700°C (middle figure) and I_s–T curves in the range −196–700°C (lower figure) for the same sample. (c) I_{rs}–T curves for ZA gabbro (upper figure) and I_s–T curve for the same sample. (d) I_{rs}–T curves for NR pillow lavas (upper figure) and for the NR dyke (lower figure). Curve 1—first heating; curve 2—second heating.

tion of pyrrhotite to magnetite. The anisotropy of susceptibility (the degree of anisotropy $P = K_1/K_3$ lies between 1.01 and 1.03 in most cases) is too low to influence the directions of NRM (Hrouda 1982). In one site representing the amphibole–pyroxene gabbro, the susceptibility was much higher ($40–100 \times 10^{-3}$SI) and its P parameter attained values of 1.14–1.27, mainly due to foliation. NRM directions from this site do not differ from those established in others. We therefore assume that they are not influenced by anisotropy.

Figure 5 shows examples of IRM acquisition curves. They reveal only a single magnetic carrier with a typical saturation

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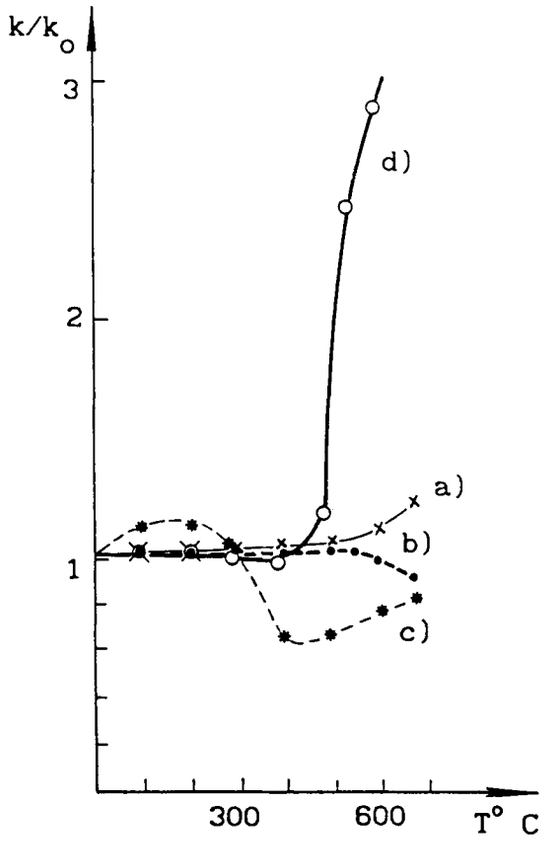


Figure 4. Susceptibility versus temperature. Plotted curves: (a) SL gabbro from zone 2; (b) SL amphibole-pyroxene gabbro; (c) SL diabase; (d) NR pillow lava.

field for magnetite. The hysteresis loops are characteristic of rocks containing small amounts of magnetically soft minerals with predominantly paramagnetic minerals.

There are many books and papers dealing with magnetic mineralogy that can be of great help in identifying magnetic minerals—carriers of rock-magnetic properties. Especially useful in our study were, for example, Haggerty (1976), Haratstra (1982), Kruczyk (1983), Kennedy & Osborne (1987), O'Reilly (1984), Dekkers (1988), Orlicky *et al.* (1992) and Tarling & Hrouda (1993). The same applies to the discussions of ZA and NR magnetic mineralogy.

The properties of the samples from the Słęża complex are summarized in Table 1.

Palaeomagnetic results

Typical results of the thermal demagnetization are presented in Figs 6(a) and (b) in the form of intensity decay curves and orthogonal vector plots. AF demagnetization was less effective than thermal demagnetization in removing the NRM components and examples of it will not be presented here.

The analyses of the demagnetization experiments performed independently in the Warsaw and Strasbourg laboratories gave similar results. Several components of NRM were isolated in three characteristic unblocking temperature (T_{ub}) ranges: 300–350°C (low-temperature component, lt), 500–600°C (high-temperature component, ht) and greater than 600°C (very high temperature component, vht). These T_{ub} ranges correspond to pyrrhotite (lt), magnetite (ht) and traces of haematite unidentified

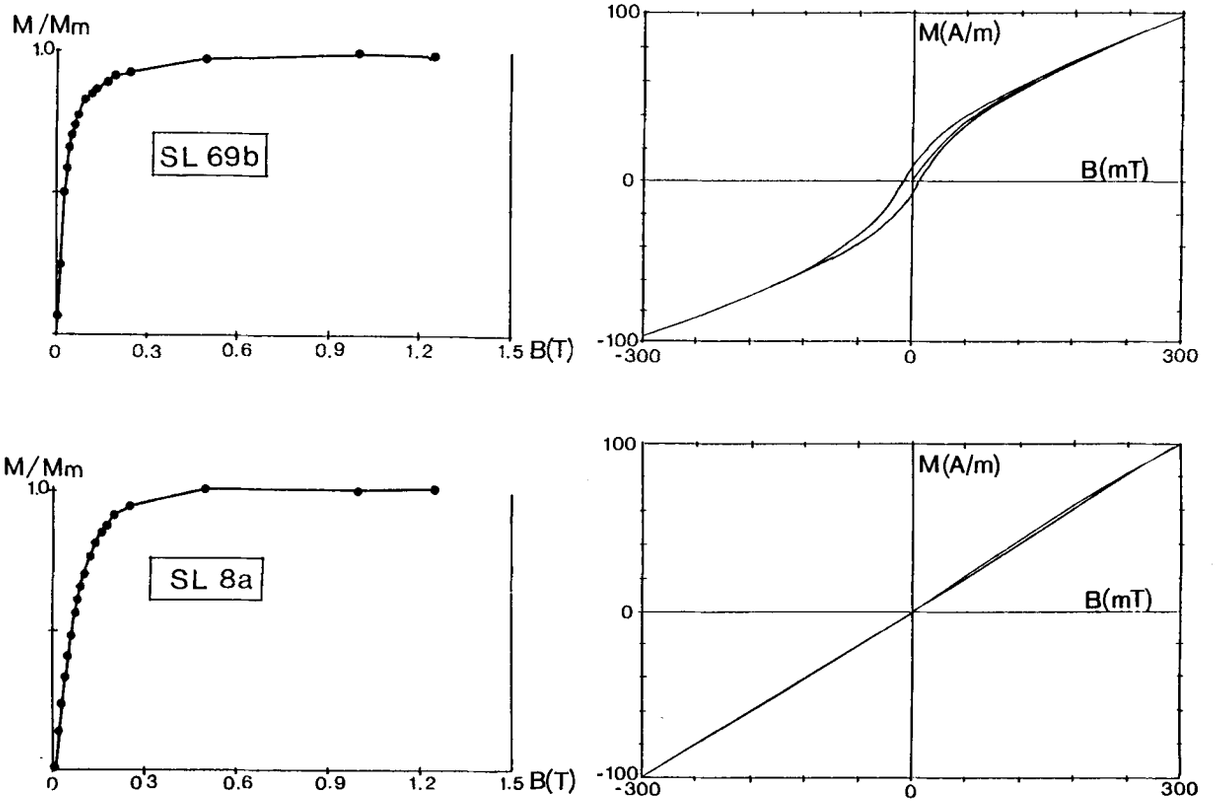


Figure 5. Normalized IRM acquisition curves and hysteresis loops for SL gabbro (two lower figures) and SL diabase (two upper figures). M—magnetization; B—field induction.

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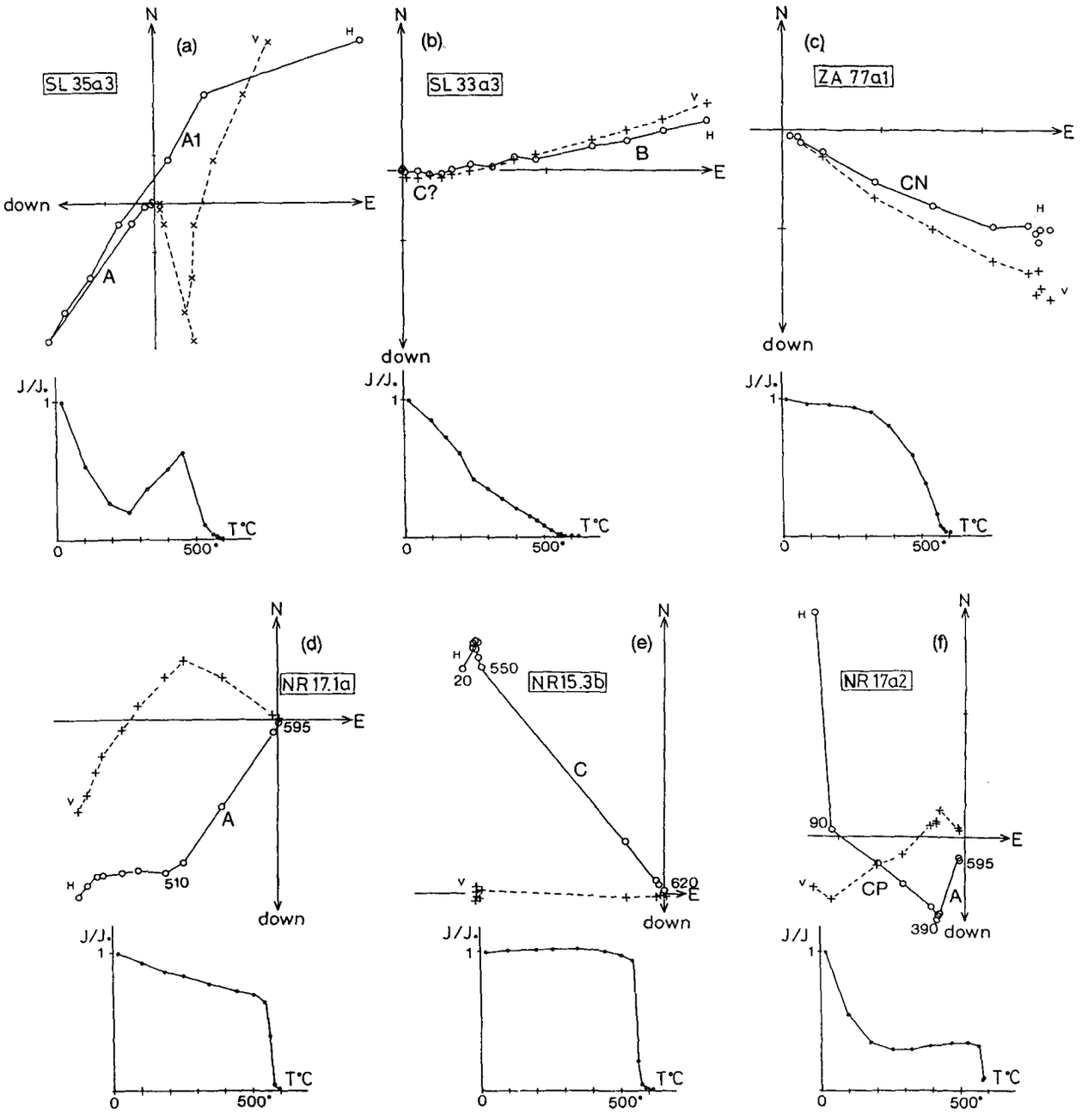


Figure 6. Results of thermal demagnetization, Zijderveld diagrams and intensity decay curves. (a) and (b) SL gabbro; (c) ZA gabbro; (d)–(f) NR gabbro. J/J_0—intensity of NRM after consecutive demagnetization steps/intensity of NRM before demagnetization. Crosses—vertical component (V); circles—horizontal component (H).

under the microscope or in the thermomagnetic curves (vht)—see Fig. 11.

Figure 7 shows the directions isolated for the five zones investigated (see Fig. 1a). In the Fe–Ti mineralized zones 1 and 2 where an ilmeno-magnetite primary assemblage prevails, the dominant directions are grouped in the declination range 180–240°C with positive and negative inclinations. In zone 2 we have also isolated directions with an upward inclination of 30–60° and a declination 120–180°C, and some with positive or negative inclinations grouped in the fourth quadrant. In zone 3 (top part of the SL unit) we distinguished the same directions as in zones 1 and 2 and, in addition, a distinct cluster of E–W directions with positive and negative shallow and intermediate inclinations. In zone 4 (edge part of the SL unit) the distribution pattern resembles that of zone 3. In zone 5 (a diabase dyke where pyrrhotite was observed as seen in Fig. 2d), only a few directions were isolated.

In order to be as objective as possible in dividing the isolated directions into groups, we used the SURFER program (Lewandowski 1992) to construct density maps of SL directions; see Fig. 8. It helped in estimating the approximate limits of the characteristic declination and inclination for each group. On this basis the mean directions

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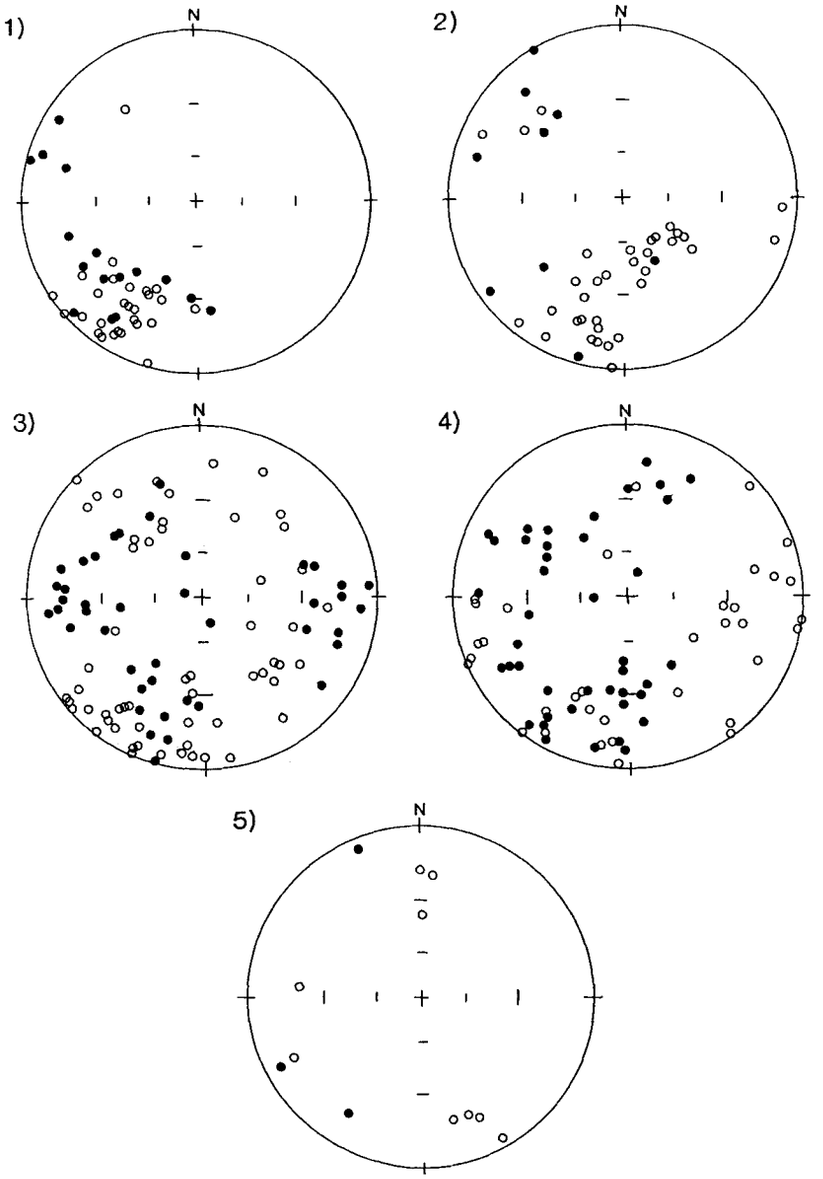


Figure 7. Stereographic projections of directions of isolated NRM components for SL zones. (1) Zone 1, (2) zone 2, (3) zone 3, (4) zone 4, and (5) zone 5. Full circles—lower hemisphere; open circles—upper hemisphere.

were calculated with the standard Fisher method for each zone. The results are presented in Fig. 9(a) and in Table 2. They resemble the directions found by Edel (1987) and Edel & Wickert (1991) for Palaeozoic rocks of the European Variscan Belt, and we have therefore called them A, A1, BW, BE, CP and CN, accordingly.

The six mean directions presented in Fig. 9(a) show that the A directions are mainly reversed. There are only six normal directions against 101 reversed. A1 and C directions have both polarities. Directions BW and BE lie near the equator—BW to the west and BE to the east—and can be regarded as one direction with both polarities, although it is difficult to say which is reversed and which is normal.

The means calculated for the normal and reversed polarity groups are compared in Fig. 10. The McFadden & McElhinny (1990) reversal test was applied to the A1, CP and B directions. According to their criteria, CP and B are classified as 'indeterminate' with critical angles γ_c equal to 25° and 21°, respectively. For the A1 direction γ_c is equal to 15°, which exceeds the observed angle between the normal and reversal means. According to the McFadden & McElhinny criteria, this direction passes the reversal test with a 'positive C' classification.

Figure 11 shows the distribution of the isolated components over the three ranges of unblocking temperatures. All of them appear in each range, but A1, BW, BE and CN and CP directions dominate in the low and very high T_{ub} ranges, i.e. ranges connected with pyrrhotite and haematite. The A directions seem to be carried by magnetite with T_{ub} lower than 600°C.

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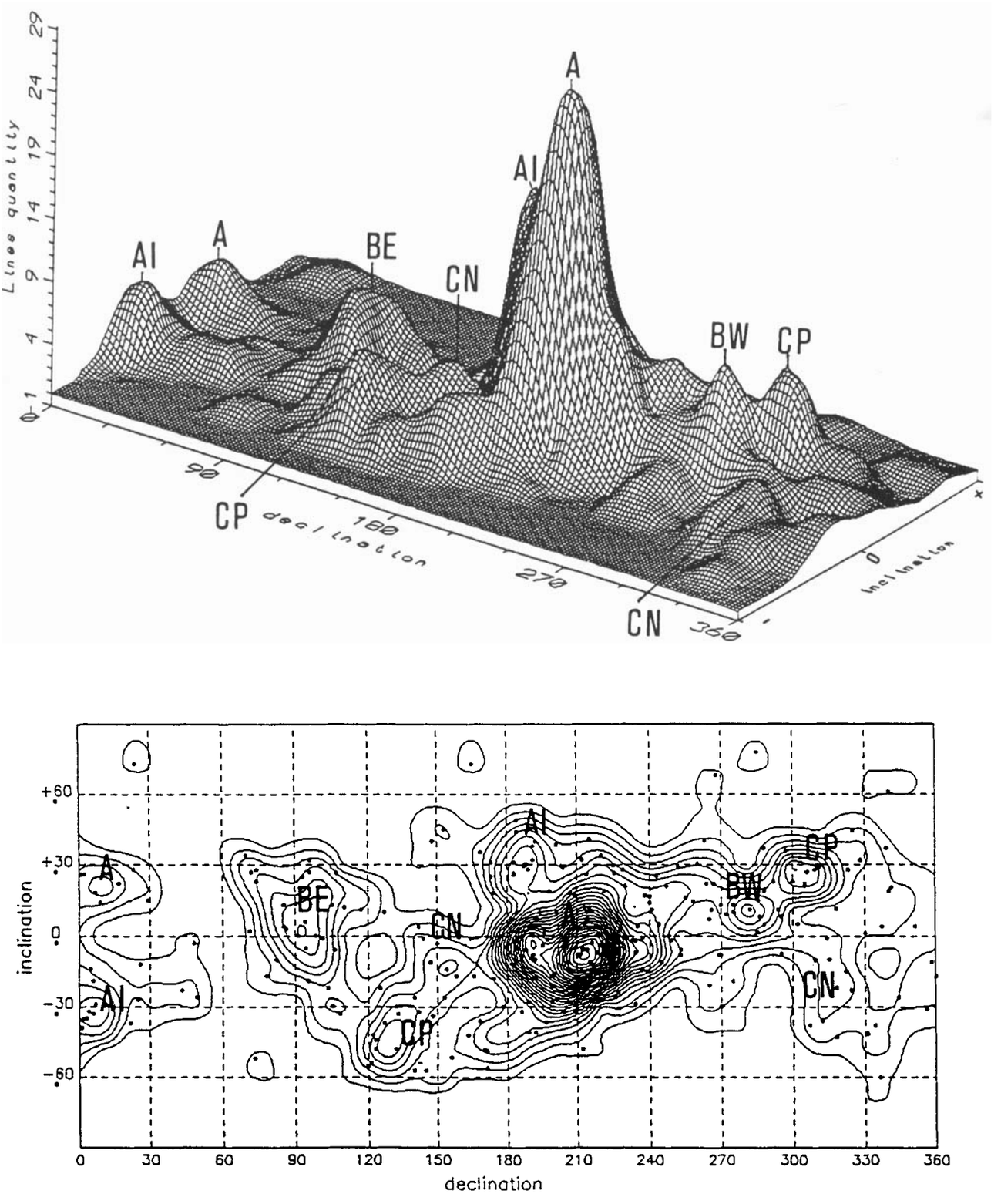


Figure 8. Density plot of all isolated directions for the SL unit, obtained with the SURFER program (Lewandowski 1992).

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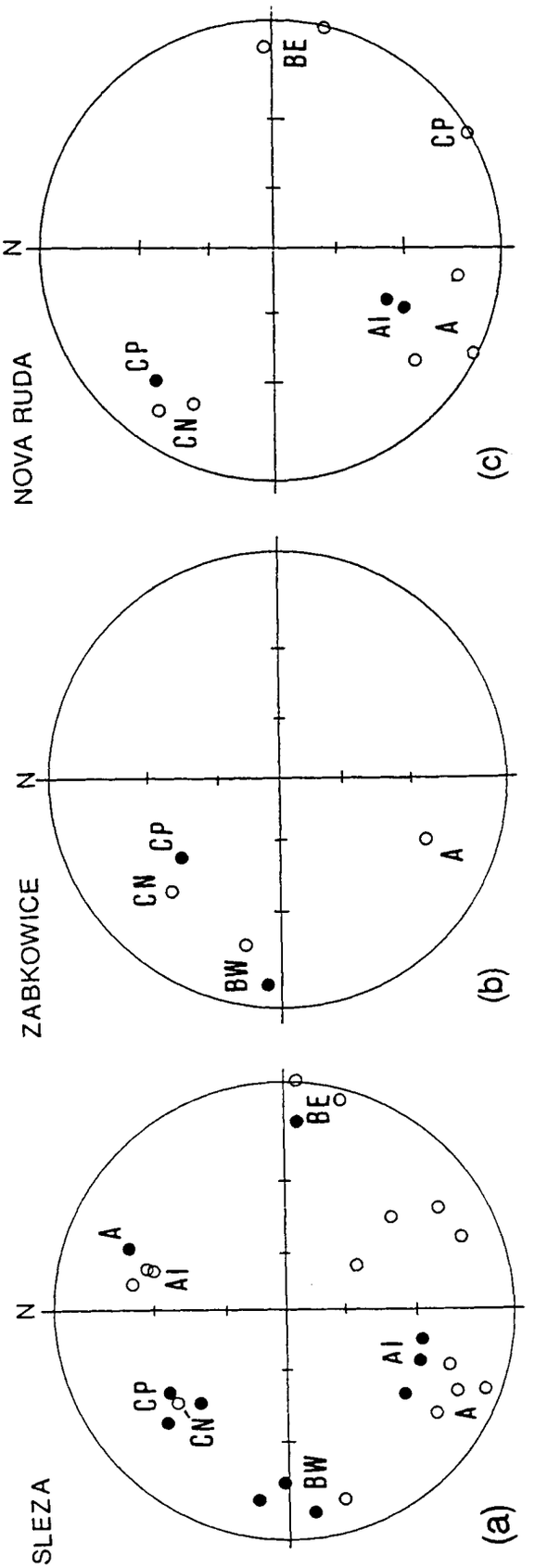


Figure 9. Stereographic projections of mean directions for the units investigated. (a) SL zones mean directions; (b) ZA mean directions; (c) NR mean directions.

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Table 2. Mean direction and pole positions obtained for the five sampling zones of Ślęża, the gabbros of Ząbkowice and the lithological units of Nowa Ruda.

Zone	S1/S2	dir	n	D	I	α95	K	LatN	LongE	Dp	Dm
Ślęża, lat = 50.90N, Long = 16.E											
1	3/11	A	31	215	-14	5	25	-37	331	3	5
		A1	10	216	26	14	13	-19	340	8	15
		BW	4	279	10	19	23	-9	103	10	20
2	3/13	A	20	199	-17	8	17	-45	350	4	8
		A1*	3	17	-26	43	9	-24	359	25	46
		BE	4	105	-3	13	53	-11	96	6	13
		CP	13	147	-50	8	29	-58	80	7	11
		CP	5	317	19	15	28	-36	72	8	15
3	4/21	A	33	206	-12	7	13	-40	342	4	7
		A1	13	201	27	8	28	-22	355	5	9
		A1	9	17	-28	12	18	-22	359	7	14
		BW	12	271	16	7	39	-7	111	4	7
		BE	14	94	11	12	12	2	100	6	12
		CP	7	138	-27	8	53	-40	75	5	9
		CP	5	325	26	15	26	-43	67	9	17
		CN	10	320	-26	11	19	-17	57	7	12
4	5/23	A	17	202	-5	8	22	-38	348	4	8
		A	6	22	18	16	18	-44	346	9	17
		A1	14	192	28	9	19	-23	344	6	17
		BW	18	263	7	12	10	2	114	6	12
		BE	12	93	0	12	14	-2	104	6	12
		CP	5	146	-13	24	11	-38	61	12	24
		CP	6	313	32	13	28	-40	83	8	15
5	1/5	A1	4	10	-22	22	19	-27	355	12	23
		B*	3	254	-9	33	15	13	116	17	33
		CP	5	157	-11	8	82	-41	48	4	9
Ząbkowice	Lat = 50.60N, Long = 16.80E										
	3/13	A	11	203	-21	14	12	-46	343	8	14
		BW	7	274	6	16	14	-5	105	8	17
		BE	8	102	17	12	24	-1	92	6	12
		CP	18	321	33	9	16	-45	75	6	10
		CN	3	314	-22	21	33	-16	64	12	23
Nowa Ruda	Lat = 50.60N, long = 16.50E										
dykes	4/18	A	7	189	-12	13	24	-45	344	7	13
		BW/BE	11	103	1	13	13	-8	96	7	13
		CP	4	150	-1	11	69	-34	54	6	11
pillow lavas	2/9	A	9	208	-1	16	11	-35	341	8	16
		A1	3	205	33	27	22	-18	352	17	30
		BW/BE	5	88	-7	18	18	-2	110	9	19
gabbro	3/15	A	10	219	14	9	27	-36	326	5	10
		A1	6	205	26	12	33	-22	350	7	13
		BE	8	86	-30	13	18	-10	120	8	15
		CP	17	311	16	10	14	-31	78	5	10
		CN	7	305	-9	12	24	-17	76	6	13
		CN	4	117	16	15	39	-10	80	8	15

S1/S2—number of sites/number of hand samples; dir—direction, n—number of directions used for calculations; D—declination in deg; I—inclination in deg; $\alpha 95$, K—Fisher confidence parameters; LatN—pole latitude; LongE—pole longitude; Dp, Dm—axes of pole confidence ellipse.
* Directions rejected from the calculation of mean directions.

Ząbkowice ZA

The gabbroic rocks of the ZA complex were subjected to the same procedure as the samples of the SL complex. Their lithology and general properties are described in Table 1.

Magnetic mineralogy and rock-magnetic characteristics

The results of optical and thermomagnetic studies of the ZA rocks show the presence of magnetite and pyrrhotite in very small amounts. In some cases fine needles dispersed inside the silicates were seen under the microscope. Thermomagnetic curves typical for most of the ZA samples are shown in Fig. 3(c).

The susceptibility values range from 200 to 1000×10^{-6} SI, and the values remain constant up to 500°C; the degree of anisotropy is very low: P does not exceed 1.02.

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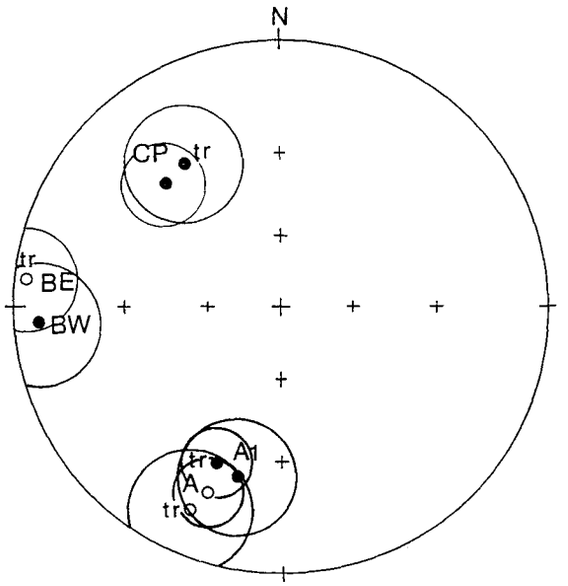


Figure 10. Stereographic distribution of mean directions calculated for normal (full circles) and reversed (open circles) polarities for the Ślęża unit. tr—directions transformed into opposite polarity for better comparison.

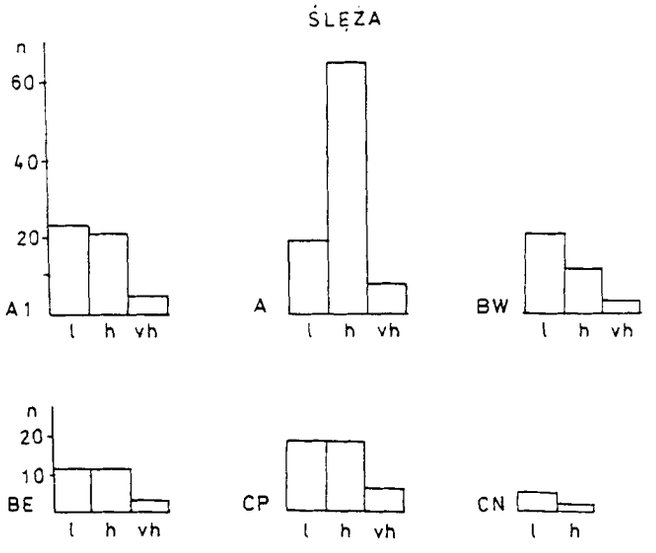


Figure 11. Histograms of isolated components for the Ślęża unit against ranges of unblocking temperatures. l—low-temperature range, h—high-temperature range, vh—very high temperature range.

Palaeomagnetic results

The standard palaeomagnetic procedure applied to the ZA rocks revealed the presence of five clusters of directions coinciding with those identified in the SL complex. The only direction not represented here is the A1 direction. Both polarities were observed here only for the B directions. Fig. 6(c) presents an example of the demagnetization results. The mean directions obtained here are shown in Fig. 9(b) and Table 2. They agree well with the results obtained for the SL complex.

Nowa Ruda, NR

This complex, although smaller than SL, is more interesting because it comprises the three rock types characteristic of an ophiolite: pillow lavas, gabbros and diabases. Its general characteristics are summarized in Table 1.

Magnetic mineralogy and rock-magnetic characteristics

The optical investigation showed the presence of small amounts of iron hydroxides of probably post-pyrite origin, some pyrrhotite, some altered ilmeno-magnetite in the gabbros, diabases and pillow lavas, and some inhomogeneously distributed magnetite in the gabbros.

Thermomagnetic curves for the pillow lavas reveal the presence of magnetite accompanied by some phases with low blocking temperatures (about 200°C) and traces of haematite. The amount of magnetic minerals increases considerably after heating; see Fig. 3(d). Thermomagnetic curves for the diabases and gabbros show the presence of magnetite and, in some cases, small amounts of pyrrhotite, as shown in Fig. 3(d).

The values of susceptibility for the pillow lavas and gabbros lie in the range $300–1000 \times 10^{-6}$ SI. The susceptibility of some diabase specimens is of the same value, while in others it reaches values of $1000–3000 \times 10^{-3}$ SI. Heating influences the susceptibility values in various ways. The susceptibility does not change at all, or only slightly, in gabbroic samples and in samples of diabases with high susceptibility. In contrast, it increases considerably in diabases with low susceptibility and in pillow lavas after heating to 475–500°C. This suggests that some of the rocks were already altered in nature. Similar changes are induced in other rocks during heating in the laboratory. Fig. 4(d) presents an example of changes of susceptibility with temperature.

The degree of anisotropy, P, does not exceed 1.04 in the lavas and dykes or 1.06 in the gabbros, with only a few exceptions.

Palaeomagnetic results

The same standard procedures as used for SL and ZA rocks were applied to the NR samples. Figs 6(d), (e) and (f) show typical demagnetization results. Analysis of the demagnetization data reveals the presence of five groups of directions corresponding to A, A1, BE, CP and CN (the three isolated BW components are enclosed within the BE group). In this unit numerous directions close to Mesozoic/Tertiary directions for Europe are observed. This indicates that the rocks were subjected to strong remagnetization at those times. The A directions are the most numerous—they appear in all three rock types. Their unblocking temperatures belong in most cases to the high-T_{ub} range, indicating that this component is related to magnetite. The A1 directions are found mainly in gabbroic samples, whereas the BE, CP and CN directions (with a predominance of CP) are present in all three rock types. Directions of both polarities appear in the A1 and CN groups. They are less easily distinguishable within the B group. All CP directions are normal. The mean directions

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are presented in Fig. 9(c) and, together with pole positions and Fisher parameters, in Table 2.

DISCUSSION OF THE RESULTS

Figure 12 shows a stereographic projection of the Fisher mean directions. As discussed earlier, the investigated complexes SL, ZA and NR are not tilted (Jamrozik 1989a), and no tectonic correction was applied to the palaeomagnetic results.

The results fall into four distinct clusters: A, A1, B and C. The group A is carried mainly by magnetite, and groups A1, B and C seem to be related mainly with pyrrhotite and, to a lesser extent, with magnetite and haematite, as was shown for SL in Fig. 11.

The directions and corresponding pole positions are summarized in Table 3. The results of Edel & Lewandowski (1993), obtained for the Variscan Belt, are also included in Table 3. This table shows that three out of the four Sudetian pole positions cited above, namely A, B and C, are in good agreement with the corresponding poles obtained by Edel & Lewandowski (1993) for the Central European Variscan Belt. The Sudetian pole A1 differs from the Variscan pole $A1_v$, but agrees well with the Silurian poles for Baltica included in Table 3 and in Fig. 13.

Our pole A is in good agreement with the results for the Permian for Baltica and for the Permian data from the Sudetes (Westphal *et al.* 1987), and also with the European poles of the same age. The negative polarity of the A directions also agrees with the polarity of the Permian Sudetian and European results. Taking this into account, we conclude that the A components record a Permo–Carboniferous field direction. They can be assigned to the low-temperature alterations of sulphides and titanomagnetites to magnetite during the period of hydrothermal and tectonic activity that lasted from the Upper Carboniferous to the Early Permian in the Sudetes (Jarmołowicz-Szulc 1987).

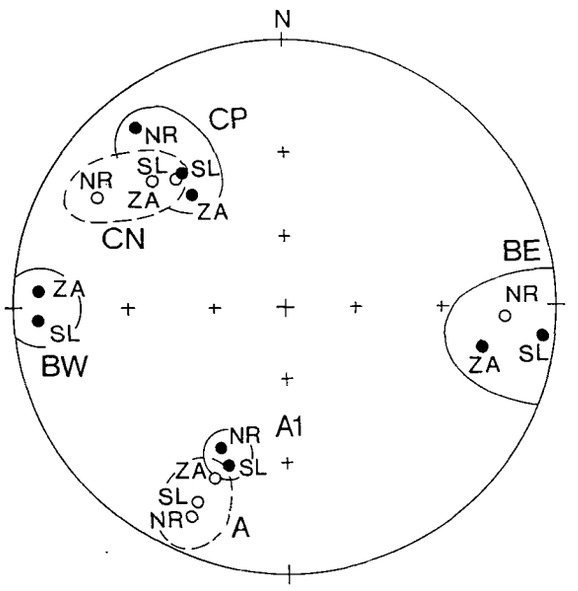


Figure 12. Mean directions for SL, ZA and NR combined.

Groups B and C are considerably shifted from the APWP for Baltica (Fig. 13). This shift cannot be explained by simple rotation of the units studied around their vertical axes, because the geological setting excludes such a possibility (Jamrozik 1989a). This conclusion is supported by our palaeomagnetic results. The mean directions obtained for the SL, ZA and NR units are very similar, although they are situated on both sides of the Marginal Sudetic Fault (SL and ZA to the north, NR to the south). This result rules out relative rotation of the massifs studied.

The problem is how to interpret the A1, B and C poles. There are two possibilities: (1) they all are overprints of fields younger than the Silurian; (2) the A1 pole is the primary Silurian pole, which implies that the B and C poles are artefacts.

Let us assume that the first idea is correct and that the A1, B and C groups record the palaeomagnetic field of a younger age than the Silurian. This hypothesis is supported by the occurrence of groups of directions very similar to B_v and C_v in the western part of the Variscan Belt (Table 3). Both Sudetic groups are carried mainly by pyrrhotite that could have been preheated above its Curie point (325°C) and partly transformed to magnetite. The remagnetization processes responsible for the B and C directions can be assigned to the two maxima of low-grade metamorphism that took place in the Sudetes of 330 ∓ 15 Ma and 300–310 Ma (Jarmołowicz-Szulc 1987). These times agree well with the ages cited in Edel & Lewandowski (1993) for Variscan Europe (see also Table 3).

If we assume that the B and C groups of remanences reflect palaeomagnetic fields and are significant, then we have to conclude that they are evidence of tectonic events that took place in the Sudetes during the Middle Carboniferous. For instance, Lewandowski (1992) explains the deviation of B_v and C_v poles from the Baltica APWP as an effect of the dextral strike-slip movements of the European Variscan Province against Baltica during the post-Early Devonian–Lower Carboniferous time-span. The situation of the B and C Sudetic poles can be explained in the same way. The appearance of both normal and reversed polarities in the B and C groups supports this explanation, although the reversal test of McFadden & McElhinny (1990), when applied to the Ślęża unit, is classified as 'indeterminate' (see above).

According to the hypothesis discussed, the A1 directions should be treated as Upper Carboniferous directions which, for some reason, differ by about 10° from the expected ones.

The alternative would be that the A1 directions are primary and represent the Silurian geomagnetic field. As a consequence, the B and C directions are artefacts.

This idea is supported by the fact that the A1 pole position agrees well with the Silurian poles for Baltica both in position and in polarity (Fig. 13, Table 3). Its mixed polarity agrees well with the results of Trench *et al.* (1993). They produce evidence to show that, during the Silurian (Wenlock excluded), multiple field reversals took place. As was shown earlier, the A1 direction for Ślęża passes the McFadden & McElhinny (1990) reversal test with a C classification. The A1 component was preserved in the less altered units of SL and NR and was absent in the more altered ZA. The abnormal B and C pole positions are situated at a great distance from the APW path for Baltica.

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Table 3. Mean directions and pole positions for the Ślęża, Ząbkowice and Nowa Ruda units.

dir	age	N	D	I	α 95	K	Lat N	Long E	Dp	Dm
Ślęża										
A	P - Cu	5	205	-13	7.5	104	-41	343	4	8
A1	S?	6	200	26	7.5	80	-23	355	4	8
A1*	S?	5	199	26	9	71	-23	357	5	10
BW	?	4	267	6	17	29	0	291	9	17
BW*	?	3	271	11	14	79	5	290	7	14
BE	?	3	97	3	15	67	4	280	8	15
CP**	?	7	323	26	11	29	42	248	7	12
CN	?	1	320	-26			17	237		
Ząbkowice										
A	P - Cu		203	-21	14	12	46	163	8	14
BW	?		274	6	16	14	-5	105	8	17
BE	?		102	17	12	24	-1	92	6	12
CP	?		321	33	9	16	-45	75	6	10
CN	?		314	-22	21	33	-16	64	12	23
Nowa Ruda										
A	P - Cu	3	205	-9	25	24	-39	343	13	26
A1	S?	2	205	30			-20	351		
BE	?	3	93	-12	28	19	6	288	15	29
CP	?	2	321	9			33	246		
CN**	?	2	301	-13			14	258		
Mean pole positions for Ślęża, Ząbkowice and Nowa Ruda									pol	Plat
A	P - Cu	3			5.5	505	-42	343	R	-7.5
A1	S?	2					-22	354	M	
A1#	S?	7			6	89	-22	355	M	14.6
BW	?	2					5	288	N	
BE	?	3			23	29	4	280	R	
BW/BE	?	5			12	40	4	283	M	0.9
CN	?	3			11	127	40	249	M	
CP	?	3			16	61	16	246	M	
CP/CN	?	6			13	27	28	248	M	1.1
Mean pole positions for Northern Vosges, Spessart, Odenwald, Pfalz according to Edel and Lewandowski (1993) denoted by suffix v, age in Ma										
Av in situ	250-270				5	75	-51	340		
A1v in situ	300-305				6	44	-23	337		
Bv in situ	307-325				8	18	8	123		
CPv in situ	325-330				5	36	41	251		
Cnv corrected	340-330				9	37	-10	46		
Cv mean							29	237		
Silurian poles for Scandinavia according to Torsvik et al., (1992)										
					9		-19	344		
					8		-23	351		
					2		-19	349		
					5		-19	352		

dir—direction; age—supposed age of magnetization; N—number of entries used for calculations; D—declination in deg; I—inclination in deg; $\alpha 95$, K—Fisher parameters of confidence; Lat N—pole latitude; Long E—pole longitude; Dp, Dm—axes of pole confidence llipse; pol—polarity: N—normal; R—reversed, M—mixed; Plat—palaeolatitude.

* Means calculated without directions marked by asterix in Table 2.

** Means calculated for normal and reversed directions together.

A1—Mean direction calculated from means for Ślęża and Nowa Ruda.

A1#—mean direction calculated for the five Ślęża zones and the Nowa Ruda pillow lavas and gabbros.

This may be the result of chemical magnetization of new phases in directions that differ from the direction of previous remanence and from the remagnetizing field direction.

As was stated previously, the rocks contain magnetic minerals of the same type but with different origins (primary and secondary magnetite, primary and secondary pyrrhotite). Therefore, one cannot exclude the second possibility that these magnetic carriers are magnetized in different directions—for example, in normal Silurian and in reversed Permian directions. In such cases, the various components would have the same spectra of blocking temperatures and, in most cases, similar spectra of coercivities. Hence, we

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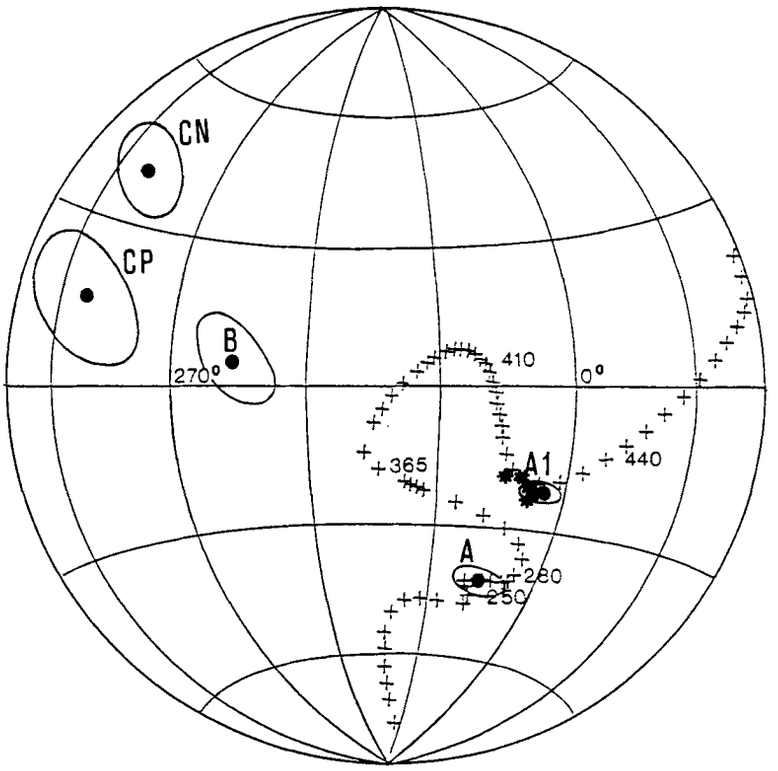


Figure 13. APWP curve for Baltica constructed by Lewandoswki (1992) after the data of Torsvik *et al.* (1992). Full circles denoted by appropriate letters are the mean directions obtained in the present study. Numbers indicate ages in Ma. Stars denote mean Silurian poles for Baltica, after Torsvik *et al.* (1992).

would measure the resultant remanence, and isolation of the different components would, in most cases, not be possible.

We have to state that to accept that A1 is of Silurian age leads to the unlikely conclusion that in the middle of the Variscan orogeny there were no tectonic disturbances in the Sudetes. The results obtained for Silurian rocks from the Bohemian Massif by Tait *et al.* (private communication) contradict this hypothesis. Their direction is closer to A than to A1 and implies a rotation of the Bohemian Massif of about 180° against Baltica.

After presenting both possible explanations of our results, we conclude that the present state of knowledge does not allow us to decide which of these two hypotheses is correct. There is further uncertainty due to the fact that a fold test is not possible.

CONCLUSIONS

Investigations of three different mafic units of Silurian age surrounding the Sowie Góry Mountains Massif result in complex but similar palaeomagnetic records with A, A1, B and C groups of palaeomagnetic directions. The complexity of the results means that the different directions might not be reliable in terms of geotectonics.

(1) The A direction is consistent with Permo–Carboniferous directions for the Sudetes, Baltica and the Variscan Belt. Overprints are assigned to low-temperature alteration processes associated with the Late Variscan hydrothermal and tectonic activity.

(2) Concerning the B and C overprints, rock magnetism studies could not provide a definite conclusion on the reliability of the directions. They may represent the magnetic field and be interpreted in terms of tectonics, as is the case for similar directions in Western Europe (Edel & Lewandowski 1993). They could also be artefacts due to remagnetization processes.

The assumption that the B and C directions are reliable and are a result of overprinting during the Sudetian phase would imply that there have been large clockwise rotations, probably related to major strike-slip motions along the south-western margin of Baltica (Badham 1982; Lewandowski 1992). In this case the A and A1 directions are the youngest post-tectonic directions.

(3) The A1 direction can be regarded either as a Carboniferous overprint or as a primary Silurian direction. If the latter is true, then B and C directions are artefacts. The implication is that the ophiolites have rested in the same relative positions against Baltica since the Silurian. There would also be no place for large oceans within the eastern part of Variscan province and the southern rim of Baltica since Silurian times.

In order to exclude one of the two hypotheses, we have to wait for more reliable data to prove or discard the idea of great strike-slip motions of Variscan Europe against Baltica.

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